MB502P – MOLECULAR BIOLOGY

Practical Exam Notes

Prepared by: *Sir Mughal*

Main Group link: https://chat.whatsapp.com/DWXwJpXahbxFVQutey8JXN

Subject Group link: https://chat.whatsapp.com/GIcr0aNQzUv23spXjzy6AM

Compiled by SIR MUGHAL

M.C.Qs (Multiple Choice Questions)

1. Which of the following step is performed to purify the RNA pellet during the RNA isolation by Trizol method? **WASHING WITH ETHANOL**
2. Which of these dyes we could use to visualize DNA in agarose gel? **ETHIDIUM BROMIDE**
3. What does a 260/280 ratio of 2.0 indicate in RNA quantification? **CONTAMINATED WITH PROTEINS**
4. The distance of DNA migration in a gel during electrophoresis primarily depends upon? **SIZE OF DNA**
5. Which component helps separate DNA from cellular proteins and other contaminants during extraction method? **PHENOL-CHLOROFORM**
6. The aqueous phase after adding phenol-chloroform and centrifugation is? **SUPERNATANT**
7. Lysis buffer used in DNA isolation is? **TRIZOL**
8. Under an electric field, in which direction DNA travels in agarose gel? **-ive TO +ive**
9. Which of the following step is NOT involved in the DNA isolation using a commercial kit? **PHENOL-CHLOROFORM SEPARATION**
10. By SDS-PAGE analysis, we can analyze? **PROTEINS, SEPARATED ON THE BASIS OF SIZE**
11. What is the function of reverse transcriptase during cDNA synthesis? **SYNTHESIZING DNA FROM RNA**
12. If we are doing the manual isolation of plasmid by alkaline lysis method, EDTA will be used in? **SOLUTION I**
13. The size of ideal primers for a PCR reaction is? **18-24 BP**
14. The purpose of the annealing step in a PCR reaction is? **TO ALLOW PRIMERS TO BIND WITH DNA**
15. The most common PCR type that is used to detect the viral load in hepatitis diagnostic tests is? **REAL-TIME PCR (qPCR)**
16. Which of the following factor determines the movement of protein in PAGE analysis? **SIZE OF THE PROTEIN**
17. Which aspect of DNA is used in RFLP analysis? **RESTRICTION OF DNA FRAGMENTS**
18. Which fluorescence method is commonly used to monitor the qRT-PCR reaction? **SYBR GREEN**
19. To make cDNA, which enzyme is used? **REVERSE TRANSCRIPTASE**
20. DNA pellet after isolation is washed with 70% ethanol to remove **SALTS & IMPURITIES.**
21. Three basic steps of conventional PCR are **DENATURATION – ANNEALING – EXTENSION**

Get in touch with us:
Sir Mughal: https://wa.me/923132310940
Sir Jawad: https://wa.me/923036100859

22. PCI is used for **DNA & RNA EXTRACTION**
23. Enzyme used in cDNA synthesis is **REVERSE TRANSCRIPTASE**
24. Which reagent is used for DNA precipitation in protein extraction? **ISOPROPANOL**
25. What is the purpose of adding chloroform to sample in RNA extraction using Trizol reagent? **TO SEPARATE RNA FROM OTHER COMPONENTS.**
26. Which enzyme is used to separate DNA from mRNA? **REVERSE TRANSCRIPTASE**

SHORT QUESTIONS

1. Why centrifugation is important and critical step in DNA isolation?

Centrifugation is a critical step in DNA isolation because it helps in the separation of cellular components based on density**.** Without centrifugation, it would be difficult to obtain pure, intact DNA because cellular debris and contaminants would remain mixed with the nucleic acids, reducing yield and quality. After cell lysis, centrifugation is used to:

- Pellet cell debris and insoluble material at the bottom of the tube.
- Separate DNA-containing supernatant from proteins, lipids, and other contaminants.
- Enable purification steps by repeatedly spinning samples after addition of reagents (e.g., phenol-chloroform or ethanol precipitation).
- Concentrate DNA by forming a visible pellet during final precipitation steps.

2. Write a short note on DNA fingerprinting.

DNA FINGERPRINTING

DNA fingerprinting is a laboratory technique used to identify individuals based on their unique DNA patterns**.** It analyzes specific regions of DNA called **v**ariable number tandem repeats (VNTRs) or short tandem repeats (STRs)**,** which vary greatly between individuals. In this method, DNA is first extracted from a sample (blood, hair, saliva, etc.) and then cut or amplified using enzymes or PCR. The resulting DNA fragments are separated using gel electrophoresis, producing a unique banding pattern for each person. These patterns are then compared between samples to determine identity or genetic relationships.

3. Write applications of DNA fingerprinting.

Applications of DNA Fingerprinting:

- ✓ ***Forensic investigations*** – Identification of criminals from biological evidence such as blood, hair, or saliva.
- ✓ ***Paternity and maternity testing*** – Determination of biological parentage.
- ✓ ***Identification of missing persons*** – Matching DNA samples with relatives.
- ✓ ***Disaster victim identification*** – Recognizing victims of accidents, wars, or natural disasters.
- ✓ ***Genetic and population studies*** – Studying genetic diversity and evolutionary relationships.
- ✓ ***Wildlife conservation*** – Identifying species and tracking endangered animal populations.

4. While running RNA, why we add reference blank sample?

While running RNA, a reference blank sample is added to serve as a control and calibration standard. The reference blank improves the accuracy and reliability of RNA quantification. It helps to:

- Set the baseline or zero absorbance of the instrument.
- Eliminate the effect of solvents, buffers, and reagents used in the sample.
- Ensure accurate measurement of RNA concentration and purity.
- Detect contamination or errors in the analysis.

Compiled by SIR MUGHAL

5. Name the reagents used for RFLP.

Reagents used for RFLP (Restriction Fragment Length Polymorphism):

- ✓ ***Restriction enzymes (restriction endonucleases)*** – to cut DNA at specific sequences.
- ✓ ***DNA sample*** – the genetic material to be analyzed.
- ✓ ***Agarose gel and electrophoresis buffer*** – for separation of DNA fragments.
- ✓ ***DNA probes (labeled probes)*** – to detect specific DNA fragments.
- ✓ ***Hybridization buffer*** – for probe binding.
- ✓ ***Southern blotting materials*** *(nylon/nitrocellulose membrane)* – for transfer of DNA fragments.

6. If do electrophoresis of a sample, is there still a need of spectrophotometer?

Yes, a spectrophotometer is still needed even after electrophoresis.

- Electrophoresis is used to assess the integrity, size, and quality of DNA/RNA fragments.
- Spectrophotometry is used to measure the concentration and purity of DNA/RNA by determining absorbance values (e.g., A260/A280 ratio).

7. Why cells are treated with $CaCl_2$ in preparation of chemically competent cell preparation?

Cells are treated with $CaCl_2$ (calcium chloride) to make them chemically competent for DNA uptake during transformation. The Ca^{2+} ions neutralize the negative charges on both the bacterial cell membrane and DNA molecules, reducing electrostatic repulsion. This allows plasmid DNA to bind more easily to the cell surface and enter the cell during heat shock. Therefore, $CaCl_2$ treatment increases the efficiency of bacterial transformation by facilitating DNA uptake.

8. What laboratory equipment are used in DNA isolation by CTAB?

Laboratory equipment used in DNA isolation by the CTAB method:

1. ***Microcentrifuge*** – for separating DNA and cellular debris.
2. ***Water bath or incubator*** – for incubation with CTAB buffer.
3. ***Micropipettes and tips*** – for accurate liquid handling.
4. ***Microcentrifuge tubes (Eppendorf tubes)*** – for sample processing.
5. ***Vortex mixer*** – for mixing reagents and samples.
6. ***Mortar and pestle (or tissue homogenizer)*** – for grinding plant tissue.
7. ***Analytical balance*** – for weighing samples and chemicals.
8. ***Spectrophotometer/NanoDrop*** – for measuring DNA concentration and purity.

9. What is Heat-Shock transformation?

Heat-shock transformation is a widely used technique for introducing foreign DNA (usually plasmid DNA) into chemically competent bacterial cells, such as *E. coli*. Before transformation, the bacterial cells are treated with calcium chloride ($CaCl_2$), which makes the cell membrane more permeable to DNA. In this method, the competent cells mixed with plasmid DNA are first kept on ice and then exposed to a brief heat shock at 42°C for about 30–90 seconds. The sudden increase in temperature creates temporary pores in the bacterial membrane, allowing the plasmid DNA to enter the cells. The cells are then returned to ice and incubated in a nutrient medium to recover. Heat-shock transformation is an important technique in genetic engineering and molecular biology because it enables the transfer of recombinant DNA into bacteria for gene cloning, protein production, and DNA amplification. It is simple, cost-effective, and commonly used in research laboratories.

Compiled by SIR MUGHAL

Get in touch with us:
Sir Mughal: https://wa.me/923132310940
Sir Jawad: https://wa.me/923036100859

10. Name the lab equipment required for DNA isolation by RFLP.

The laboratory equipment commonly used in RFLP analysis includes:

- Microcentrifuge for DNA extraction and purification.
- Micropipettes and sterile tips for accurate handling of samples and reagents.
- Water bath or incubator for restriction enzyme digestion.
- Agarose gel electrophoresis unit for separating DNA fragments.
- Power supply to run electrophoresis.
- UV transilluminator or gel documentation system for visualizing DNA bands.
- Southern blotting apparatus for transferring DNA fragments onto a membrane.
- Hybridization oven/incubator for probe hybridization.
- Spectrophotometer or NanoDrop for determining DNA concentration and purity.

11. Why blood sample is preferred for DNA isolation?

Blood is preferred for DNA isolation because it is one of the most reliable and convenient sources of genomic DNA. It contains white blood cells (leukocytes), which are nucleated cells carrying the complete genetic material of an individual, making DNA extraction efficient and high yielding. Blood is also easy to collect using minimally invasive methods, such as venipuncture, and can be obtained in relatively large quantities compared to other tissues. When properly stored with anticoagulants, blood DNA remains stable and less prone to degradation. Additionally, blood samples are widely used in forensic investigations, paternity testing, and genetic studies because they provide consistent and reproducible results across different individuals and conditions.

12. Write steps for making competent bacterial cells.

Steps for making competent bacterial cells

- ✓ ***Selection of bacterial culture:*** A fresh colony of *E. coli* is inoculated in LB broth and grown until it reaches mid-log phase ($OD_{600} \approx 0.4–0.6$)**,** when cells are most active and suitable for transformation.
- ✓ ***Cooling the culture:*** The bacterial culture is kept on **ice** to slow metabolism and stabilize the cell membrane.
- ✓ ***Harvesting cells:*** Cells are collected by centrifugation, forming a pellet while the supernatant is discarded.
- ✓ ***Washing with $CaCl_2$ solution:*** The cell pellet is gently resuspended in ice-cold calcium chloride ($CaCl_2$)**.** This step is repeated 1–2 times to remove contaminants.
- ✓ ***Membrane permeabilization:*** Ca^{2+} ions neutralize negative charges on the bacterial membrane and DNA, making the cells competent for DNA uptake.
- ✓ ***Incubation on ice:*** Cells are kept on ice for a period to increase membrane stability and prepare them for transformation.
- ✓ ***Storage or immediate use:*** Competent cells are either used immediately for transformation or stored at -80°C for future use.

13. Write disadvantages of PCR.

Disadvantages of PCR:

- ***High sensitivity to contamination:*** Even a small amount of contaminating DNA can lead to false results.
- ***Requires prior sequence knowledge:*** Primers must be designed based on known DNA sequences.
- ***Cannot distinguish live vs dead organisms:*** It amplifies DNA regardless of whether the source is viable.
- ***Risk of amplification errors:*** DNA polymerase may introduce mutations during amplification.
- ***Limited fragment size:*** Usually works best for short DNA fragments; long sequences are difficult to amplify.
- ***Requires expensive equipment and reagents:*** Thermocycler, primers, and enzymes increase cost.
- ***False positives/negatives possible:*** Due to technical errors, poor primer design, or inhibitors in the sample.

Compiled by: SIR MUGHAL

Get in touch with us:
Sir Mughal: https://wa.me/923132310940
Sir Jawad: https://wa.me/923036100859

14. Write steps involved in organic DNA extraction.

Steps involved in organic DNA extraction:

- ✓ ***Sample collection and cell lysis:*** The biological sample (blood, tissue, or plant material) is collected and treated with a lysis buffer (detergent + salt) to break cell and nuclear membranes and release DNA.
- ✓ ***Removal of proteins and contaminants:*** Organic solvents such as phenol and chloroform are added to denature proteins and separate them from nucleic acids.
- ✓ ***Phase separation by centrifugation:*** The mixture is centrifuged, forming:
    1. Upper aqueous phase (DNA)
    2. Interphase (proteins/debris)
    3. Lower organic phase (lipids and denatured proteins)

- ✓ ***Collection of DNA-containing layer:*** The upper aqueous layer containing DNA is carefully transferred to a fresh tube.
- ✓ ***DNA precipitation:*** Cold ethanol or isopropanol is added to precipitate DNA as visible strands or pellets.
- ✓ ***DNA washing:*** The DNA pellet is washed with 70% ethanol to remove residual salts and impurities.
- ✓ ***DNA rehydration (dissolving):*** The purified DNA is dissolved in TE buffer or sterile water for storage and further analysis.

15. Write roles of Ethidium bromide, TAE buffer and Bromophenol blue.

Ethidium Bromide (EtBr):

Ethidium bromide is a fluorescent dye used in gel electrophoresis for the visualization of DNA. It intercalates between the base pairs of DNA and emits bright orange fluorescence when exposed to ultraviolet (UV) light. This property allows researchers to detect and analyze DNA bands after electrophoretic separation. Although highly effective, it is also mutagenic, so it must be handled with care.

TAE Buffer (Tris–Acetate–EDTA):

TAE buffer is commonly used as a running buffer in agarose gel electrophoresis. It provides a stable pH and maintains proper ionic strength, which is essential for the smooth migration of DNA fragments through the gel. The Tris component maintains pH stability, acetate provides conductivity, and EDTA chelates divalent metal ions like Mg^{2+}, thereby protecting DNA from degradation by nucleases.

Bromophenol Blue:

Bromophenol blue is a tracking dye used in gel electrophoresis as part of the loading dye mixture. It helps researchers monitor the progress of DNA migration through the gel by moving ahead of most DNA fragments. It also increases the density of the sample, ensuring that it sinks properly into the gel wells during loading, allowing accurate and easy sample application.

16. Differentiate between SYBR Green and Taqman probe.

SYBR Green:

SYBR Green is a double-stranded DNA binding dye used in real-time PCR. It binds non-specifically to any double-stranded DNA and fluoresces upon binding. This means it detects all amplified products, including non-specific products like primer dimers. It is simple, cost-effective, and does not require probe design, but it has lower specificity compared to probe-based methods.

TaqMan Probe:

TaqMan probe is a sequence-specific fluorescent probe used in real-time PCR. It contains a reporter dye and a quencher dye; during PCR amplification, the probe is cleaved by Taq polymerase, separating the dyes and producing fluorescence. It detects only the target-specific DNA sequence, making it highly specific and accurate, but more expensive due to probe design and synthesis.

Compiled by: SIR MUGHAL

Get in touch with us:
Sir Mughal: https://wa.me/923132310940
Sir Jawad: https://wa.me/923036100859

17. What components are needed in cDNA synthesis?

cDNA synthesis requires specific enzymes and reagents to convert mRNA into complementary DNA (cDNA). The main components include:

- ***mRNA template:*** The starting material containing the expressed gene sequence.
- ***Reverse transcriptase enzyme:*** Catalyzes synthesis of DNA from RNA template.
- ***Oligo(dT) primers / gene-specific primers:*** Initiate DNA synthesis by binding to RNA.
- ***dNTPs (deoxynucleotide triphosphates):*** Building blocks for new DNA strand synthesis.
- ***Buffer solution:*** Maintains optimal pH and ionic conditions for enzyme activity.
- ***Mg^{2+} ions:*** Act as cofactors for reverse transcriptase activity.
- ***RNase inhibitor:*** Prevents degradation of RNA during the reaction.

18. Why water is not used in electrophoresis?

Water is not used in electrophoresis because it cannot provide proper ionic conductivity and buffering capacity. Electrophoresis requires a buffer system (like TAE or TBE) to maintain a stable pH and supply ions that carry electric current through the gel. Pure water has very low ion concentration, so it cannot support efficient movement of DNA or proteins. In addition, without a buffer, the pH would fluctuate during the run due to electrolysis, which can damage samples and affect their migration. Therefore, buffer solutions are used instead of water to ensure stable pH, proper electrical conductivity, and accurate separation of biomolecules.

19. How to prepare 1X TAE buffer from 50X TAE buffer?

To prepare 1X TAE buffer from 50X TAE stock solution, a simple dilution is performed. Use the formula:

$$C_1V_1 = C_2V_2$$

Where:

- C_1 = 50X (stock concentration)
- C_2 = 1X (final concentration)

So, 1 part of 50X TAE is diluted with 49 parts of distilled water.

Steps:

- Measure the required volume of 50X TAE buffer (e.g., 20 mL).
- Add distilled water to make the final volume 50 times dilution (e.g., 980 mL water for 20 mL stock).
- Mix well to obtain 1X TAE working solution.

20. Write a short note on SDS-PAGE.

SDS-PAGE (Sodium Dodecyl Sulfate–Polyacrylamide Gel Electrophoresis):

SDS-PAGE is a widely used analytical technique for the separation of proteins based on their molecular weight. In this method, proteins are first treated with SDS (an anionic detergent), which denatures them and gives a uniform negative charge proportional to their length, eliminating the effect of shape and native charge. The protein samples are then loaded onto a polyacrylamide gel, and an electric field is applied. Smaller proteins move faster through the gel matrix, while larger proteins move more slowly, allowing separation according to size. After electrophoresis, proteins are visualized using stains such as Coomassie Brilliant Blue or silver stain. SDS-PAGE is commonly used in protein analysis, purification studies, and molecular weight determination.

Compiled by: SIR MUGHAL

Get in touch with us:
Sir Mughal: https://wa.me/923132310940
Sir Jawad: https://wa.me/923036100859

21. Write a short note on silica based DNA isolation.

Silica-based DNA isolation:

Silica-based DNA isolation is a widely used method in which DNA binds selectively to a silica membrane or silica particles in the presence of chaotropic salts such as guanidinium chloride. During cell lysis, DNA is released and mixed with binding buffer, allowing it to attach to silica under high-salt conditions while proteins and other contaminants do not bind. After binding, the silica matrix is washed with ethanol-based buffers to remove impurities, salts, and proteins. Finally, DNA is eluted using low-salt buffer or nuclease-free water, which releases purified DNA from the silica surface. This method is fast, reliable, and commonly used in commercial DNA extraction kits due to its high purity and efficiency.

22. What will happen if we use water instead of buffer in gel preparation and gel tank for gel electrophoresis?

If water is used instead of buffer in gel preparation and the electrophoresis tank, the process will not work properly.

- ***No proper current flow:*** Water has very low ion concentration, so it cannot conduct electricity efficiently, resulting in weak or no migration of DNA/proteins.
- ***No pH stability:*** Without buffer (TAE/TBE), pH changes during electrophoresis due to electrolysis, which affects the stability and movement of biomolecules.
- ***Poor or no separation:*** DNA bands may become smeared or may not separate at all due to inconsistent electric field.
- ***Damage to samples:*** Uncontrolled pH and heat generation can degrade DNA or proteins.

23. Why we use SDS and ethanol?

SDS (Sodium Dodecyl Sulfate):

SDS is an anionic detergent used in molecular biology mainly for protein denaturation and cell lysis. It breaks lipid membranes and unfolds proteins by disrupting non-covalent bonds, giving proteins a uniform negative charge. This is essential in techniques like SDS-PAGE where proteins are separated based on size rather than shape or charge.

Ethanol:

Ethanol is commonly used in DNA/RNA procedures for nucleic acid precipitation and washing. In the presence of salts, ethanol reduces DNA solubility, causing it to aggregate and form a visible pellet. It is also used in 70% form for washing DNA pellets to remove salts, enzymes, and other impurities while keeping DNA intact.

24. What would be the alternative if spectrophotometer is not available in lab?

If a spectrophotometer is not available, DNA/RNA concentration can be estimated using the following alternatives:

- ***Agarose gel electrophoresis with known DNA ladder:*** Sample band intensity is compared with standard DNA markers to approximate concentration.
- ***Fluorometer (e.g., using fluorescent dyes like SYBR Green or PicoGreen):*** Gives more sensitive and accurate quantification than gel estimation.
- ***Colorimetric methods:*** DNA reacts with specific dyes (e.g., diphenylamine or orcinol-based assays) to produce color intensity proportional to concentration.
- ***Visual estimation (crude method):*** Comparing pellet size or band brightness, though this is least accurate.

Compiled by: SIR MUGHAL

Get in touch with us:
Sir Mughal: https://wa.me/923132310940
Sir Jawad: https://wa.me/923036100859

25. Why TE buffer is used in DNA extraction instead of water?

TE buffer is used in DNA extraction instead of water because it provides protection and stability to DNA. TE buffer contains Tris (Tris base) which maintains a stable pH, preventing acid or alkaline conditions that can degrade DNA. It also contains EDTA (Ethylenediaminetetraacetic acid), which chelates divalent metal ions like Mg^{2+} and Ca^{2+}. These ions are required for DNase activity, so EDTA inhibits nucleases and protects DNA from enzymatic degradation. In contrast, water does not maintain pH or inhibit nucleases, so DNA becomes more prone to degradation and instability. Therefore, TE buffer ensures long-term preservation and integrity of extracted DNA.

26. What is the role of PCI, EDTA buffer and Ethanol?

PCI (Phenol:Chloroform:Isoamyl alcohol):

PCI is used in DNA extraction for removal of proteins and other contaminants. Phenol and chloroform denature proteins and separate them from nucleic acids, while isoamyl alcohol reduces foaming and improves phase separation. After centrifugation, DNA remains in the aqueous phase.

EDTA buffer:

EDTA acts as a chelating agent that binds divalent metal ions like Mg^{2+} and Ca^{2+}. These ions are essential cofactors for DNase enzymes, so EDTA inhibits nuclease activity and protects DNA from degradation. It also helps maintain DNA stability during storage.

Ethanol:

Ethanol is used for DNA precipitation and purification. In the presence of salts, it reduces DNA solubility, causing DNA to aggregate and form a visible pellet during centrifugation. It is also used in 70% form to wash DNA pellets, removing salts and impurities while keeping DNA intact.

27. Write three applications of RFLP.

Applications of RFLP:

- ***Forensic science:*** Used for DNA profiling to identify criminals from biological evidence such as blood, hair, or saliva.
- ***Paternity testing:*** Helps determine biological relationships by comparing DNA band patterns between parents and offspring.
- ***Genetic disease diagnosis:*** Used to detect mutations or gene variations associated with inherited disorders.

28. What is the purpose of stock solutions? Name three stock solutions that are used for isolation of DNA.

Purpose of stock solutions:

Stock solutions are highly concentrated, pre-prepared solutions of reagents that are stored and later diluted to required working concentrations. They ensure accuracy, consistency, time efficiency, and reproducibility in experiments by reducing repeated preparation errors.

Three stock solutions used for DNA isolation:

- ***TE buffer (Tris-EDTA stock solution)*** – used to maintain DNA stability and protect it from degradation.
- ***Tris buffer stock solution*** – maintains pH during DNA extraction.
- ***EDTA solution*** – chelates metal ions and inhibits DNase activity to protect DNA.

Compiled by: SIR MUGHAL

29. Write short note of SYBR Green dye.

SYBR Green dye:

SYBR Green is a fluorescent dye used in real-time PCR (qPCR) for detecting double-stranded DNA. It binds specifically to the minor groove of double-stranded DNA and emits strong fluorescence when bound, which increases proportionally with DNA amplification. During PCR, as the amount of amplified DNA increases, fluorescence intensity also increases, allowing real-time monitoring of DNA amplification. It is a simple and cost-effective method because it does not require sequence-specific probes. However, SYBR Green is non-specific, meaning it binds to any double-stranded DNA, including non-specific products and primer dimers, which can sometimes affect accuracy.

30. Write steps of plasmid isolation using alkaline lysis.

Steps of plasmid isolation using alkaline lysis:

- ✓ ***Cell harvesting:*** Bacterial culture containing plasmid is centrifuged to collect the cell pellet.
- ✓ ***Resuspension:*** The pellet is resuspended in Tris-EDTA buffer containing RNase to stabilize cells and degrade RNA.
- ✓ ***Cell lysis (alkaline lysis):*** A solution containing SDS and NaOH is added. SDS disrupts cell membranes and proteins, while NaOH denatures chromosomal and plasmid DNA.
- ✓ ***Neutralization:*** Potassium acetate solution is added to neutralize the mixture. This causes chromosomal DNA, proteins, and cell debris to precipitate, while plasmid DNA remains in solution due to its small, circular nature.
- ✓ ***Centrifugation:*** The mixture is centrifuged to separate insoluble debris (pellet) from plasmid-containing supernatant.
- ✓ ***DNA precipitation:*** The supernatant is treated with cold ethanol or isopropanol to precipitate plasmid DNA.
- ✓ ***Washing and resuspension:*** The DNA pellet is washed with 70% ethanol, dried, and finally dissolved in TE buffer or sterile water for further use.

31. What is supernatant? What care should be taken when removing it from sample?

Supernatant:

Supernatant is the clear liquid layer that remains above the pellet after centrifugation. It contains dissolved components such as DNA, proteins, salts, or other biomolecules depending on the procedure, while heavier debris forms the pellet at the bottom.

Care while removing supernatant:

- Do not disturb or resuspend the pellet, as it may contain target material (e.g., DNA or cells).
- Use a micropipette or sterile pipette tip carefully and slowly.
- Tilt the tube slightly and remove liquid from the opposite side of the pellet.
- Avoid aspirating too close to the bottom to prevent contamination.
- Change tips between samples to avoid cross-contamination.

Compiled by: SIR MUGHAL

Get in touch with us:
Sir Mughal: https://wa.me/923132310940
Sir Jawad: https://wa.me/923036100859

LONG QUESTIONS

1. Differentiate between PCR and qRT-PCR.

Difference between PCR and qRT-PCR

Feature	PCR (Polymerase Chain Reaction)	qRT-PCR (Quantitative Reverse Transcription PCR)
Definition	Technique used to amplify a specific DNA segment	Technique that converts RNA into cDNA and amplifies it with real-time quantification
Starting material	DNA	RNA (converted into cDNA)
Main enzymes	Taq DNA polymerase	Reverse transcriptase + Taq DNA polymerase
Process type	Single-step DNA amplification	Two-step: reverse transcription + real-time PCR
Detection method	End-point analysis (after PCR completion)	Real-time detection during amplification
Quantification	Qualitative or semi-quantitative	Fully quantitative (measures gene expression levels)
Detection tools	Agarose gel electrophoresis	Fluorescent dyes (SYBR Green) or probes (TaqMan)
Sensitivity	Moderate	High sensitivity
Applications	DNA cloning, mutation detection, gene identification	Gene expression analysis, viral load detection, diagnostics
Result output	Band on gel	Amplification curve (Ct value)

2. Write steps of RT-PCR.

Steps of RT-PCR (Reverse Transcription PCR):

- ✓ ***RNA extraction:*** Total RNA is isolated from the biological sample (cells, blood, tissue, or virus-infected material) using appropriate extraction methods. High-quality, intact RNA is essential for reliable results.
- ✓ ***Reverse transcription (cDNA synthesis):*** The extracted RNA is converted into complementary DNA (cDNA) using the enzyme reverse transcriptase. Primers such as oligo(dT), random hexamers, or gene-specific primers are used to initiate synthesis.
- ✓ ***Preparation of PCR mixture:*** The cDNA is mixed with Taq DNA polymerase, primers (forward and reverse), dNTPs, $MgCl_2$, and buffer solution to prepare the reaction mixture.
- ✓ ***Denaturation step:*** The mixture is heated (usually ~94–95°C) to separate double-stranded DNA into single strands.
- ✓ ***Annealing step:*** Temperature is lowered (around 50–65°C) to allow primers to bind to complementary sequences on the template DNA.
- ✓ ***Extension/Elongation:*** Taq polymerase extends the primers at ~72°C, synthesizing new DNA strands.
- ✓ ***Amplification cycles:*** Steps 4–6 are repeated for 25–40 cycles, resulting in exponential amplification of the target cDNA.
- ✓ ***Detection and analysis:*** Amplified products are analyzed either by gel electrophoresis (end-point RT-PCR) or fluorescence measurement (qRT-PCR) depending on the type of RT-PCR used.

Compiled by: SIR MUGHAL

Get in touch with us:
Sir Mughal: https://wa.me/923132310940
Sir Jawad: https://wa.me/923036100859

3. Write principle and procedure of Gel Electrophoresis.

Principle of Gel Electrophoresis:

Gel electrophoresis is based on the movement of charged biomolecules (mainly DNA, RNA, or proteins) in an electric field through a porous gel matrix. DNA is negatively charged due to its phosphate backbone, so when an electric current is applied, it migrates towards the positive electrode (anode). The gel (agarose or polyacrylamide) acts as a molecular sieve, separating molecules based on their size and charge, where smaller fragments move faster and travel farther than larger ones.

Procedure of Gel Electrophoresis:

- ✓ ***Gel preparation:*** Agarose powder is dissolved in TAE or TBE buffer, heated, and poured into a casting tray with a comb to form wells. After solidification, the gel is placed in an electrophoresis tank.
- ✓ ***Sample preparation:*** DNA samples are mixed with loading dye (e.g., bromophenol blue) which adds density and helps track movement.
- ✓ ***Loading samples:*** Prepared samples and a DNA ladder (marker) are carefully loaded into the wells using a micropipette.
- ✓ ***Electrophoresis run:*** The tank is filled with buffer, and an electric current is applied. DNA fragments migrate through the gel towards the anode.
- ✓ ***Separation of fragments:*** Smaller DNA fragments move faster through the gel pores, while larger fragments move slowly, resulting in separation based on size.
- ✓ ***Staining and visualization:*** After running, DNA bands are stained using ethidium bromide or safe alternatives and visualized under a UV transilluminator.
- ✓ ***Result analysis:*** Band patterns are compared with the DNA ladder to estimate fragment size and quantity.

4. Write steps involved in plasmid isolation.

Steps involved in plasmid isolation:

- ✓ ***Inoculation and culture growth:*** Bacterial cells (e.g., *E. coli*) containing plasmid are grown in LB broth with appropriate antibiotic selection to ensure plasmid retention.
- ✓ ***Cell harvesting:***The bacterial culture is centrifuged to collect the cell pellet containing plasmid-bearing cells.
- ✓ ***Resuspension:*** The pellet is resuspended in Tris-EDTA buffer containing RNase to stabilize cells and degrade RNA.
- ✓ ***Alkaline lysis:*** A lysis solution containing SDS and NaOH is added. SDS disrupts cell membranes and proteins, while NaOH denatures both chromosomal and plasmid DNA.
- ✓ ***Neutralization:*** Potassium acetate (or acidic buffer) is added to neutralize the mixture. This causes chromosomal DNA, proteins, and cell debris to precipitate, while **ci**rcular plasmid DNA remains in solution due to its small size and supercoiled nature.
- ✓ ***Centrifugation:*** The mixture is centrifuged to separate the precipitated debris (pellet) from the plasmid-containing supernatant.
- ✓ ***Plasmid DNA precipitation:*** The supernatant is treated with cold isopropanol or ethanol to precipitate plasmid DNA.
- ✓ ***Washing:*** The DNA pellet is washed with 70% ethanol to remove salts and impurities.
- ✓ ***Drying and resuspension:*** The pellet is air-dried and dissolved in TE buffer or nuclease-free water for storage and further analysis.

Compiled by SIR MUGHAL

Get in touch with us:
Sir Mughal: https://wa.me/923132310940
Sir Jawad: https://wa.me/923036100859

5. Write precautions while isolating RNA by Trizol method.

Precautions while isolating RNA by TRIzol method:

- ✓ ***Maintain RNase-free conditions:*** Use RNase-free tubes, tips, and reagents. Wear gloves at all times to prevent RNA degradation by RNases.
- ✓ ***Avoid RNase contamination:*** Clean work surfaces and equipment with RNase decontamination solutions or 70% ethanol before starting.
- ✓ ***Work quickly and on ice:*** RNA is highly unstable, so samples should be processed rapidly and kept cold wherever possible.
- ✓ ***Proper phase separation:*** After adding chloroform, ensure correct centrifugation so that clear separation of aqueous and organic phases is obtained; only the aqueous phase contains RNA.
- ✓ ***Careful handling of aqueous phase:*** Avoid disturbing the interphase and organic layer while transferring RNA-containing supernatant.
- ✓ ***Use proper centrifugation conditions:*** Maintain recommended speed and time to ensure efficient separation of phases.
- ✓ ***Avoid repeated freeze-thaw cycles:*** RNA samples should be stored properly at -80°C to prevent degradation.
- ✓ ***Use fresh reagents:*** TRIzol and other chemicals should be within expiry and properly stored for consistent results.

6. Write brief description of steps of RFLP for DNA extraction to analysis.

Steps of RFLP (from DNA extraction to analysis):

- ✓ ***DNA extraction:*** Genomic DNA is isolated from biological samples (blood, tissue, etc.) using standard extraction methods to obtain pure DNA.
- ✓ ***Restriction digestion:*** The purified DNA is cut using specific restriction endonucleases, which recognize and cleave DNA at particular sequences, producing fragments of varying lengths.
- ✓ ***Gel electrophoresis:*** The digested DNA fragments are separated on an agarose gel under an electric field. Fragments are separated based on size, with smaller fragments moving faster.
- ✓ ***Denaturation of DNA:*** The separated double-stranded DNA in the gel is treated (usually with alkali) to convert it into single-stranded DNA for probe binding.
- ✓ ***Southern blotting:*** DNA fragments are transferred from the gel onto a nylon or nitrocellulose membrane to make them stable for hybridization.
- ✓ ***Probe hybridization:*** A labeled DNA probe complementary to the target sequence is added. It binds specifically to matching DNA fragments.
- ✓ ***Detection and analysis:*** The bound probe is detected using radioactive, fluorescent, or chemiluminescent methods, producing a banding pattern (DNA fingerprint) that is analyzed for comparison.

Compiled by: SIR MUGHAL

Get in touch with us:
Sir Mughal: https://wa.me/923132310940
Sir Jawad: https://wa.me/923036100859

7. How DNA is isolated from mammalian tissue?

DNA isolation from mammalian tissue:

- ✓ ***Sample collection and homogenization:*** Mammalian tissue is collected and mechanically homogenized (using mortar and pestle or homogenizer) in a suitable lysis buffer to break tissue structure and release cells.
- ✓ ***Cell lysis:*** The homogenate is treated with detergents (e.g., SDS) and salts to break cell and nuclear membranes, releasing DNA into solution.
- ✓ ***Protein digestion:*** Proteinase K is added to degrade proteins such as histones and nucleases, ensuring clean DNA release.
- ✓ ***Removal of contaminants:*** Organic extraction using phenol-chloroform (PCI method) or other purification methods separates proteins and lipids from DNA.
- ✓ ***Phase separation by centrifugation:*** Centrifugation separates the mixture into Upper aqueous phase (DNA), Interphase (proteins), Organic phase (lipids/debris)
- ✓ ***DNA precipitation:*** The aqueous phase is treated with cold ethanol or isopropanol, causing DNA to precipitate as visible strands or a pellet.
- ✓ ***Washing:*** The DNA pellet is washed with 70% ethanol to remove salts and impurities.
- ✓ ***Drying and rehydration:*** The pellet is air-dried and dissolved in TE buffer or nuclease-free water for storage and downstream applications.

8. Explain working principle of Nano drop spectrophotometer. Evaluate how it works to identify the quantity of the sample.

Working principle of NanoDrop spectrophotometer:

NanoDrop spectrophotometer works on the principle of UV–Visible light absorbance spectrophotometry, specifically based on the ability of nucleic acids and proteins to absorb light at specific wavelengths. It uses the Beer–Lambert Law, which states that absorbance is directly proportional to the concentration of the analyte in the sample. Unlike traditional spectrophotometers, NanoDrop uses a very small sample volume (1–2 µL) placed between two optical fiber surfaces. The instrument forms a liquid column between these surfaces and passes UV light through it. The absorbance is measured at specific wavelengths, mainly:

- 260 nm for DNA/RNA
- 280 nm for proteins
- 230 nm for organic contaminants

Working for quantification of sample:

1. A small droplet of sample is placed on the measurement pedestal.
2. The instrument automatically forms a liquid column between optical fibers.
3. UV light passes through the sample, and absorbance is recorded.
4. The NanoDrop calculates concentration using:
   - DNA: A260 × 50 µg/mL (double-stranded DNA standard factor)
   - RNA: A260 × 40 µg/mL
5. Purity is assessed using ratios:
   - A260/A280 (protein contamination check; ideal ~1.8 for DNA)
   - A260/A230 (organic salt contamination check)

Evaluation (how it identifies quantity)

NanoDrop provides rapid, accurate, and direct quantification of nucleic acids using absorbance without requiring cuvettes or large sample volumes. It determines concentration by measuring how much UV light is absorbed at 260 nm and converting it into concentration using standard conversion factors based on the Beer–Lambert law. This makes it highly efficient for routine DNA/RNA quality and quantity assessment in molecular biology labs.

Compiled by: SIR MUGHAL

Get in touch with us:
Sir Mughal: https://wa.me/923132310940
Sir Jawad: https://wa.me/923036100859

9. What factors you will consider when primer designing is done.

Factors to consider in primer designing:

- ✓ ***Primer length:*** Ideal length is 18–25 nucleotides. This ensures specific binding to the target sequence.
- ✓ ***GC content:*** Should be around 40–60%**.** Proper GC content provides stable primer binding.
- ✓ ***Melting temperature (Tm):*** Forward and reverse primers should have similar Tm (usually 55–65°C) to ensure efficient annealing.
- ✓ ***Specificity:*** Primers must bind only to the target sequence and should not show significant homology with non-target regions.
- ✓ ***GC clamp:*** Presence of 1–2 G or C bases at the 3' end helps in stable binding and efficient extension.
- ✓ ***Avoid secondary structures:*** Primers should not form hairpins, self-dimers, or cross-dimers, as these reduce amplification efficiency.
- ✓ ***Amplicon size:*** For standard PCR, product size should typically be 100–1000 bp depending on application.
- ✓ ***3' end stability:*** The 3' end is crucial for extension; it should be free from mismatches or repeats.
- ✓ ***Avoid repetitive sequences:*** Long repeats or runs of a single nucleotide should be avoided to prevent non-specific binding.
- ✓ ***No complementarity between primer pairs:*** Forward and reverse primers should not anneal to each other to avoid primer-dimer formation.

10. How DNA is isolated from plant tissues?

DNA isolation from plant tissues:

- ✓ ***Sample collection and grinding:*** Fresh plant tissue is collected and ground into a fine powder using a mortar and pestle (often with liquid nitrogen) to break tough cell walls and prevent DNA degradation.

- ✓ ***Cell lysis:*** The powdered tissue is mixed with a **C**TAB extraction buffer (containing detergent, salt, and Tris buffer) to break cell membranes and release cellular contents.

- ✓ ***Removal of contaminants:*** Plant tissues contain high amounts of polysaccharides and polyphenols, so CTAB helps remove them. Sometimes PVP (polyvinylpyrrolidone) is added to bind polyphenols.

- ✓ ***Protein digestion:*** Proteinase K or other enzymes may be used to digest proteins and histones associated with DNA.

- ✓ ***Phase separation (organic extraction):*** The mixture is treated with chloroform:isoamyl alcohol (CIA) to separate DNA from proteins and other cellular debris.

- ✓ ***Centrifugation:*** After centrifugation, DNA remains in the upper aqueous phase, while proteins and debris form lower phases.

- ✓ ***DNA precipitation:*** DNA is precipitated by adding cold isopropanol or ethanol, forming visible strands or a pellet.

- ✓ ***Washing:*** The DNA pellet is washed with 70% ethanol to remove salts and impurities.

- ✓ ***Drying and resuspension:*** The purified DNA is air-dried and dissolved in TE buffer or nuclease-free water for storage and further use.

Compiled by: SIR MUGHAL

Get in touch with us:
Sir Mughal: https://wa.me/923132310940
Sir Jawad: https://wa.me/923036100859

11. Write functions of the following: Taq polymerase, dNTPs, Primers

Taq DNA Polymerase

Taq polymerase is a thermostable DNA polymerase enzyme isolated from *Thermus aquaticus*. Its main function is to synthesize new DNA strands during PCR by adding nucleotides to the 3′ end of primers. It remains active at high temperatures (~72°C), which is essential during the denaturation and extension cycles of PCR, making DNA amplification possible.

dNTPs (deoxynucleotide triphosphates)

dNTPs are the building blocks of DNA synthesis, including dATP, dTTP, dCTP, and dGTP. During PCR, Taq polymerase incorporates these nucleotides into the growing DNA strand according to the template sequence. They also provide the energy required for bond formation during DNA strand elongation.

Primers

Primers are short single-stranded DNA sequences that provide a starting point for DNA synthesis. They bind to complementary regions on the template DNA and define the specific target region to be amplified. Both forward and reverse primers are required to initiate DNA synthesis in PCR.

12. Enlist the reagent names with their uses or purpose for organic DNA extraction.

Reagents used in organic DNA extraction and their purposes:

- ✓ ***Lysis buffer (Tris + SDS):*** Breaks open cell and nuclear membranes and releases DNA into solution.
- ✓ ***EDTA (Ethylenediaminetetraacetic acid):*** Chelates Mg^{2+} ions and inhibits DNase enzymes, protecting DNA from degradation.
- ✓ ***NaCl (Salt solution):*** Helps in DNA stabilization and separation of proteins from nucleic acids.
- ✓ ***Proteinase K:*** Digests proteins, especially histones and nucleases, to release pure DNA.
- ✓ ***Phenol:Chloroform:Isoamyl alcohol (PCI):*** Denatures and removes proteins and lipids; separates them from DNA during phase separation.
- ✓ ***RNase (optional):*** Removes RNA contamination to obtain pure DNA.
- ✓ ***Cold ethanol / isopropanol:*** Precipitates DNA by reducing its solubility.
- ✓ ***70% ethanol:*** Washes DNA pellet and removes salts and impurities without dissolving DNA.
- ✓ ***TE buffer (Tris-EDTA):*** Dissolves and stores purified DNA while maintaining stability and preventing degradation.

Compiled by: SIR MUGHAL

Get in touch with us:
Sir Mughal: https://wa.me/923132310940
Sir Jawad: https://wa.me/923036100859

13. Write a detailed note on extraction of bacterial cells.

Extraction of bacterial cells (bacterial cell harvest and isolation):

Bacterial cell extraction refers to the process of obtaining bacterial cells from a culture medium for downstream applications such as DNA/RNA isolation, protein extraction, or transformation studies. The procedure is mainly based on cell harvesting and separation of cells from growth media.

1. ***Culturing of bacteria:*** Bacteria (commonly *E. coli*) are first grown in a suitable liquid medium such as Luria-Bertani (LB) broth under optimal conditions (temperature, pH, and shaking). Cells are allowed to grow until they reach the log phase or stationary phase, depending on experimental requirements, where sufficient biomass is available.

2. ***Harvesting of cells:*** Once the desired growth is achieved, the culture is transferred into sterile centrifuge tubes and subjected to centrifugation (usually 5,000–10,000 rpm for 5–10 minutes). Due to centrifugal force, bacterial cells settle at the bottom forming a cell pellet, while the liquid medium (supernatant) remains above.

3. ***Removal of supernatant:*** The supernatant containing spent media and waste products is carefully discarded without disturbing the pellet. This step is critical to avoid loss of bacterial cells.

4. ***Washing of cells:*** The pellet may be washed with sterile phosphate buffer saline (PBS) or TE buffer to remove residual media components. The cells are again centrifuged to obtain a clean pellet.

5. ***Resuspension of cells:*** The bacterial pellet is resuspended in an appropriate buffer depending on downstream use. For example, TE buffer for DNA work or lysis buffer for protein/DNA extraction. This step ensures uniform suspension of cells for further processing.

6. ***Storage or immediate use:*** The extracted bacterial cells can either be used immediately for experiments (e.g., plasmid isolation, transformation) or stored at low temperatures (4°C for short-term or -80°C for long-term storage after appropriate treatment).

14. Describe the protocol of isolation of plasmid through silica column.

Protocol for plasmid isolation using silica column method:

- ✓ ***Bacterial culture and harvesting:*** Bacterial cells containing plasmid DNA are grown in selective media (with antibiotic) and harvested by centrifugation to obtain a cell pellet.
- ✓ ***Resuspension:*** The pellet is resuspended in resuspension buffer (Tris-EDTA + RNase A) to maintain pH and degrade RNA.
- ✓ ***Cell lysis:*** Lysis buffer (SDS + NaOH or chaotropic salts in kit-based systems) is added to break cell membranes and release cellular contents, including plasmid and chromosomal DNA.
- ✓ ***Neutralization:*** A neutralization buffer (commonly potassium acetate) is added. This precipitates chromosomal DNA, proteins, and debris, while small circular plasmid DNA remains in solution.
- ✓ ***Centrifugation:*** The mixture is centrifuged to separate insoluble debris (pellet) from the plasmid-containing clear supernatant.
- ✓ ***Binding to silica column:*** The supernatant is mixed with a high-salt binding buffer (chaotropic salts) and transferred to a silica spin column. Under these conditions, plasmid DNA binds specifically to the silica membrane**.**
- ✓ ***Washing:*** The column is washed with ethanol-based wash buffers to remove salts, proteins, and other contaminants while DNA remains bound to silica.
- ✓ ***Drying spin:*** A brief centrifugation step removes residual ethanol, preventing interference in downstream applications.
- ✓ ***Elution:*** Plasmid DNA is eluted using low-salt buffer (TE buffer or nuclease-free water)**,** which disrupts DNA–silica interaction and releases purified plasmid DNA.

Compiled by: SIR MUGHAL

Get in touch with us:
Sir Mughal: https://wa.me/923132310940
Sir Jawad: https://wa.me/923036100859

15. Write mechanism of action of SYBR Green and how its fluorescence visualize the sample?

Mechanism of action of SYBR Green and fluorescence-based visualization

SYBR Green is a fluorescent intercalating dye used in real-time PCR (qPCR) to detect and quantify double-stranded DNA (dsDNA). Its mechanism is based on its ability to bind selectively to dsDNA and emit fluorescence upon excitation.

Mechanism of action:

- ✓ ***Binding to DNA:*** SYBR Green dye binds in the minor groove of double-stranded DNA by intercalating between base pairs. It has very low fluorescence in free (unbound) form.

- ✓ ***Fluorescence activation:*** When SYBR Green is bound to dsDNA, its structure becomes restricted, which significantly increases its ability to emit fluorescence upon excitation with blue light (~497 nm).

- ✓ ***Signal generation during PCR:*** During each PCR cycle, as DNA is amplified, more dsDNA is formed. SYBR Green binds to these newly formed strands, resulting in a progressive increase in fluorescence intensity proportional to DNA quantity.

- ✓ ***Real-time detection:*** The qPCR machine measures fluorescence at each cycle. The increase in signal is plotted as an amplification curve, allowing quantification of starting DNA concentration based on Ct value.

Visualization of sample using SYBR Green fluorescence:

- The instrument excites SYBR Green-bound DNA using a specific wavelength of light.
- The dye emits bright green fluorescence (~520 nm emission) when bound to dsDNA.
- The detector captures this emitted light and converts it into fluorescence intensity data.
- Higher fluorescence = higher amount of amplified DNA.
- This allows real-time monitoring and comparison of DNA amplification between samples.

Compiled by: SIR MUGHAL

Get in touch with us:
Sir Mughal: https://wa.me/923132310940
Sir Jawad: https://wa.me/923036100859

LMS Handling

by Sir Mughal

Reliable • Professional • Result-Oriented



Our Key Features

- Timely Activities
- Expert Team Members
- Maximum Quiz Scores
- Assignment Solutions by *Subject-Oriented Experts*
- Paid Project Handling
- Guaranteed Results
- Market Competitive Rates

Contact Us

0313-2310940

shahzeb.zoologist04@gmail.com